

MATH 213 Quiz 3

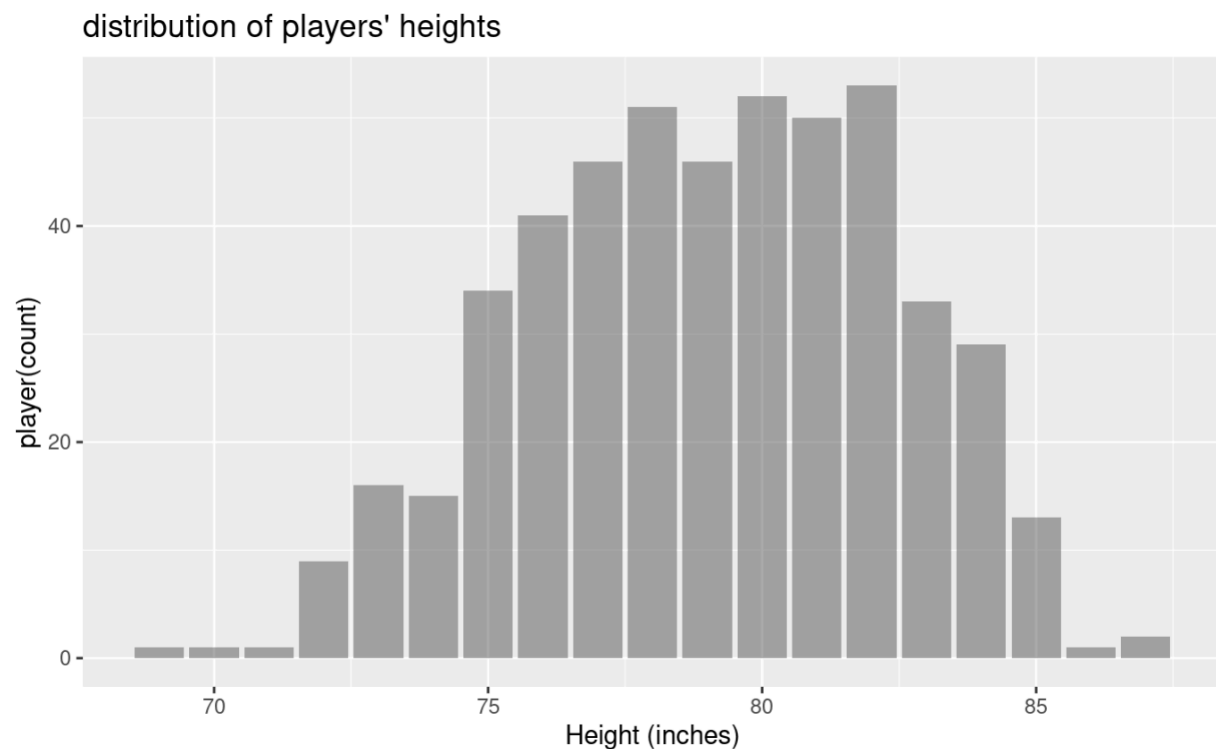
Load the data file called `nba\_players\_19.csv` from our textbook's web site. Here is the web address for the dataset: https://www.openintro.org/data/csv/nba_players_19.csv We store the data file under the name `players`.

You will find the **codebook** for the data at

https://www.openintro.org/data/index.php?data=nba_players_19

Explore the dataset in the usual ways -- **view the dataset, read the codebook!**

1. After making several graphs, paste your favorite one here that describes the heights of ALL players.



2. Paste a screenshot of the statistics you calculated.

Mean height: 79.05263

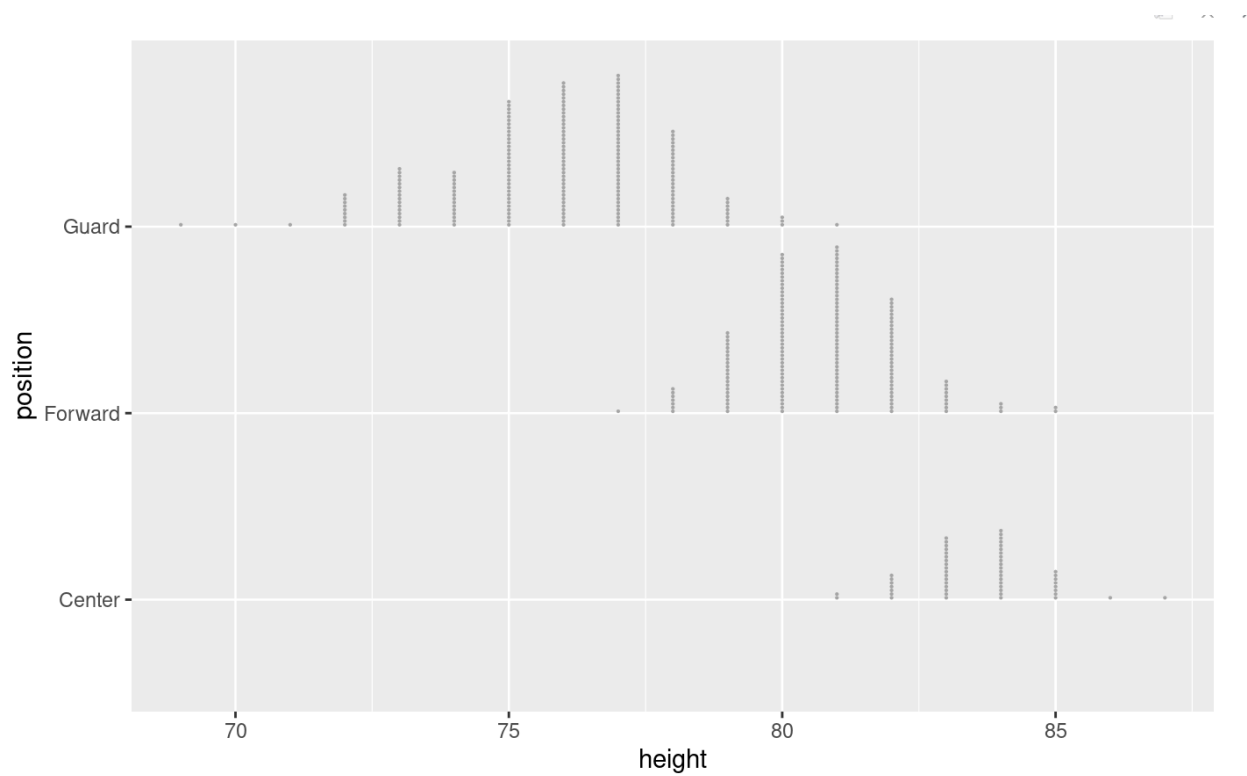
Median height: 79

Sd height: 3.31284

3. Describe the distribution of heights, addressing the center, spread, and shape. Use the context of the data set throughout your description. (NBA players from 2018-19 season, heights in inches, etc.) Remember to support your claims with the statistics you have calculated.

The data shows the height of all NBA players in the 2018-19 NBA season. The average height of an NBA player playing in this season is 79.05 inches, and the height ranges from 70 inches to 90 inches. Almost all the heights of the players are in between the lengths of 72 to 85 inches. The shape of the graph is bell shaped which means most of the NBA players are at the height of 80 inches. There are some outliers that are taller than the average, their heights are 87 inches and 89 inches tall, there are also players that are well below the average at the height of 65 inches.

4. Paste a copy of your graph here.



5. Paste a screenshot of the statistics you calculated.

R console

3 x 9

A tibble: 3 x 9

position <chr>	q025 <dbl>	q25 <dbl>	q50 <dbl>	q75 <dbl>	q975 <dbl>	mean_height <dbl>	median_height <int>	sd_height <dbl>
Center	81.35	83	84	84	85.65	83.56364	84	1.166883
Forward	78.00	80	81	82	83.95	80.71779	81	1.412070
Guard	72.00	75	76	77	79.00	75.83590	76	2.000965

3 rows

6. Write a few sentences discussing what you learn about the heights of NBA players from your graph. Remember to compare the groups on each key component of the distributions. Include some specific numbers to support your observations.

The graph shows the heights of the position of guard, forward, and center in the NBA. The graph shows the difference between heights. The guards are between the heights of 72 and 79 inches with the median of the heights being 77 inches. For the forwards the heights range from 78 to 84 inches with its median 81 inches. The center's heights are ranging from 81 to 84 inches. With this data each position has their own average height with guards being 77 inches on average, forwards being 81 inches, and centers being 84 inches. The guard's data has the most variation for the heights compared to the other two positions, shown by the standard deviation for the guards is the largest being 2.009 and the forwards being 1.412 and the centers being 1.1668